

Experiment 4: Design Zero-shot, One-shot, and Few-shot Prompts to Generate a Professional Email Requesting Leave Due to Illness

Aim

To implement Zero-shot, One-shot, and Few-shot prompting techniques using the OpenAI API to generate a professional leave request email due to illness and compare the generated emails based on tone, grammar, formatting, and completeness.

Python Code:

```
from openai import OpenAI
```

```
# Paste your OpenAI API Key here
```

```
client = OpenAI(  
    api_key="sk-proj-  
fXs9TwmdiZQU9td92kDIe9wACeeq8S0wsiGwQRfrbEoJq4VGPLU1Np74KCilODIH8_C  
oBWZp8nT3BlbkFJBjvFWrf8sOhvLZ7zO3YVr2MDh2NeIM42jru5-  
z2nctrTGHuEgdMAYbFx2lo34b4CY-q8ffMBkA"  
)
```

```
# ----- Zero-shot Prompt -----
```

```
zero_shot = """
```

```
Write a professional email requesting leave due to illness.
```

```
"""
```

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=zero_shot  
)
```

```
print("===== ZERO-SHOT EMAIL =====")
```

```
print(response.output_text)
```

```
# ----- One-shot Prompt -----
```

```
one_shot = """
```

```
Example:
```

```
Subject: Leave Request
```

```
Dear Manager,
```

```
I am feeling unwell and have been advised to take rest. I kindly request leave for today.  
I will complete any pending work after returning.
```

Thank you for your understanding.

Sincerely,
John

Now write a similar professional email requesting leave due to illness.

"""

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=one_shot  
)
```

```
print("\n===== ONE-SHOT EMAIL =====")  
print(response.output_text)
```

```
# ----- Few-shot Prompt -----  
few_shot = """  
Example 1
```

Subject: Sick Leave Request

Dear Sir,

I am suffering from a fever and will be unable to attend work today.
Kindly grant me one day of leave.

Thank you.

Regards,
David

Example 2

Subject: Medical Leave Request

Dear Madam,

Due to illness, I am unable to perform my duties today.
Please approve my leave for one day.

Thank you for your consideration.

Sincerely,

Emily

Now write a professional email requesting leave due to illness.

"""

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=few_shot  
)
```

```
print("\n===== FEW-SHOT EMAIL =====")  
print(response.output_text)
```

Output

===== **ZERO-SHOT EMAIL** =====

Subject: Leave Request Due to Illness

Dear [Manager's Name],

I am writing to inform you that I am unwell and need to take sick leave from [start date] to [end date], if necessary. I have consulted/plan to consult a doctor and will keep you updated on my condition.

I will ensure that any urgent tasks are handed over appropriately or addressed as soon as possible upon my return. Please let me know if you need any further information.

Thank you for your understanding.

Sincerely,
[Your Name]

===== **ONE-SHOT EMAIL** =====

Subject: Leave Request

Dear Manager,

I am not feeling well and have been advised to take rest. I kindly request leave for today due to illness. I will ensure that any pending work is completed once I return.

Thank you for your understanding.

Sincerely,
[Your Name]

===== **FEW-SHOT EMAIL** =====

****Subject:**** Leave Request Due to Illness

Dear Sir/Madam,

I am writing to inform you that I am unwell and unable to attend work today. I kindly request leave for one day due to illness.

I apologize for any inconvenience caused and appreciate your understanding.

Thank you.

Regards,
[Your Name]

Result

The Zero-shot, One-shot, and Few-shot prompting techniques were successfully implemented using the OpenAI API to generate a professional leave request email due to illness. The generated emails were compared based on tone, grammar, formatting, and completeness. The Few-shot prompt produced the most professional, well-structured, and complete email among the three prompting techniques.